

2 Fig. 2.1 shows three molecules of water.

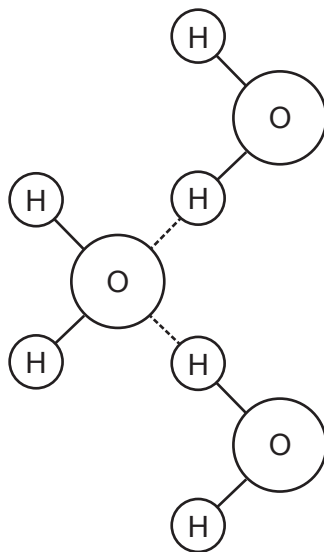


Fig. 2.1

(a) Describe the hydrogen bonding that occurs between the water molecules shown in Fig. 2.1.

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(b) The human enzyme, salivary amylase, is composed of one polypeptide. Fig. 2.2 represents the structure of a molecule of salivary amylase.

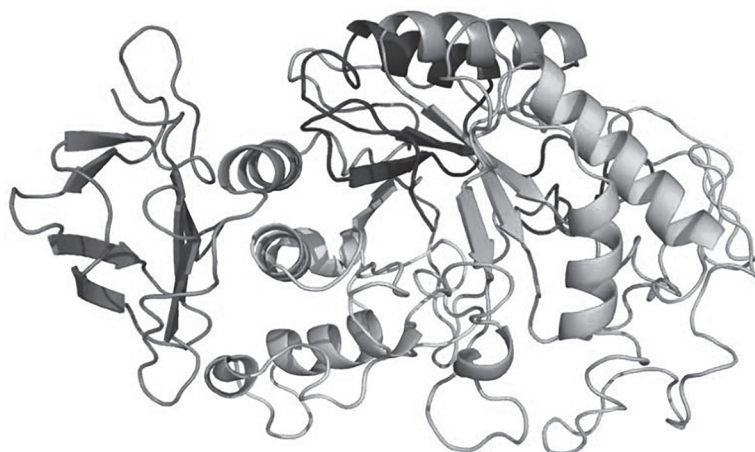


Fig. 2.2

- (i) Explain the role of hydrogen bonding in maintaining the secondary structure of proteins, such as salivary amylase.

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- (ii) Explain the role of hydrogen bonding in maintaining the tertiary structure of proteins such as salivary amylase.

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- (c) Outline the importance of water as a solvent in **plants**.

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[Total: 9]